Hawkes Process Estimation on High-Frequency Cryptocurrency Data via Neural Networks

From recurrent LSTMs to attention-based architectures

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Context

Context — why do prices move?

Volatility *is* activity [Bouchaud et al., 2018]

Minimal microstructure model: the price moves by $\pm\vartheta$ (one tick) at a rate ϕ per unit time (the market **activity**), with up and down moves equally likely. The price volatility is then

$$\boxed{\sigma^2 = \phi\,\vartheta^2}$$ (Bouchaud et al., 2018, Eq. 2.18).

- To first order, **volatility is the rate ϕ of price-change events**.
- Modelling volatility $\equiv$ modelling the **intensity** ϕ of a point process.

Consequence

If ϕ varies randomly in time, then **volatility is stochastic** — with its own dynamics.

Context — volatility is stochastic and clustered

The empirical fact

Activity (hence volatility) is **intermittent**: it alternates calm and bursts, with **long memory** (*volatility clustering*) [Bouchaud et al., 2018].

Point-process formalism

- $N(t) = \sum_i \mathbf{1}_{t_i \le t}$ counts events; intensity $\phi(t) = \mathbb{E}[\mathrm{d}N(t)]/\mathrm{d}t$.
- **Clustering ratio**: $r(\tau) = \dfrac{\mathbb{V}[N(t+\tau) - N(t)]}{\mathbb{E}[N(t+\tau) - N(t)]}$.
- Inter-event durations: cf. ACD models [Pacurar, 2006].

Poisson is not enough

A homogeneous Poisson process (ϕ constant) gives $r = 1$ (no clustering). In markets, $r \gg 1$: the intensity $\phi = \phi(t)$ must be **stochastic**.

Context — the Hawkes process

The intensity becomes its own model

[Hawkes, 1971; intro: Muni Toke, 2011]

$$\phi(t) = \underbrace{\phi_0(t)}_{\text{exogenous}} + \underbrace{\sum_{t_i<t} \Phi(t-t_i)}_{\text{endogenous (self-excitation)}}$$

- Each event raises ϕ by a kernel $\Phi \geq 0$ that decays in time.
- As $\phi(t)$ depends on **its own history**, it is *itself stochastic*.
- With $\sigma^2(t) = \vartheta^2 \phi(t)$, **volatility becomes stochastic and clustered** [Bouchaud et al., 2018].

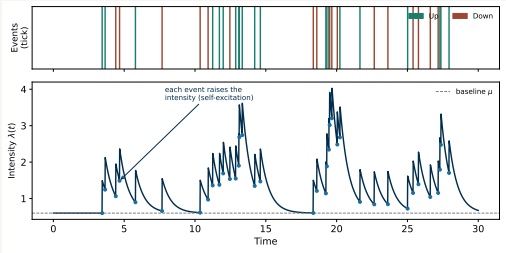


Figure: Simulated Hawkes process: events (top) raise the intensity $\phi(t)$ (bottom), which decays until the next one.

Context — endogeneity and reflexivity

Branching ratio g

$$g = \int_0^\infty \Phi(u)\,\mathrm{d}u < 1 \;\Rightarrow\; \bar{\phi} = \frac{\phi_0}{1-g}, \quad \bar{S} = \frac{1}{1-g}.$$

- Branching view: g = expected number of "child" events each event triggers (mother $\rightarrow$ children).
- $g \to 1$: **near-critical** market, activity mostly **endogenous** (*reflexivity*) [Hardiman, Bercot & Bouchaud, 2013].
- The process memory grows as $1/(1-g)$.

In our model

Exponential kernel $\Phi(t) = \alpha\, e^{-\beta t}$:

$$g = \alpha/\beta.$$

Estimating $\{\mu, \alpha, \beta\}$ = measuring **baseline**, **feedback strength** and its **memory**.

Why it matters

Estimating Hawkes $\equiv$ measuring, interpretably, the market's **volatility** and its **degree of endogeneity**.

Motivation

Motivation

- Hawkes estimation is traditionally done by **maximum likelihood (MLE)**: accurate and well-founded, but **iterative and costly**, sensitive to the initial guess and hard to use in real time.
- **Lee (2023)** [Lee, 2023] proposes estimating the parameters with an **LSTM**: the network learns on simulated data and predicts the parameters directly — accuracy comparable to MLE, inference time $< 1\%$ of MLE.
- Lee's study uses **US equities (2018–2019)**. The method is promising, but **little explored** in other markets and with **other network architectures**.

Gap

There is a lack of studies that (i) apply neural-network Hawkes estimation to **high-frequency cryptocurrency data** and (ii) investigate whether **attention-based architectures** improve estimation over the reference LSTM.

Motivation — why cryptocurrencies?

A fertile, challenging ground

- A **24/7**, liquid market with **public tick data** (e.g. exchange order-book and trade dumps) — no cost for equity NBBO data.
- **Clustered jumps and heavy tails**: Hawkes outperforms Poisson for jumps in crypto [Taddeo, Salvi & Rivera-Castro, 2024].
- An emerging market in Brazil, still little studied under this lens [Primo, 2018].

Opportunity

The more “excited” dynamics of crypto make it an ideal case to test whether **attention** models capture long-range dependencies better.

Objectives and Hypothesis

Hypothesis and Objective

Research hypothesis

A neural network can estimate the parameters of a marked Hawkes process from high-frequency cryptocurrency data with **accuracy comparable to maximum likelihood at orders-of-magnitude lower computational cost**; moreover, **attention-based architectures** can **match or surpass** the reference LSTM in estimation quality, especially under crypto's more clustered, heavy-tailed jump dynamics.

Main objective

Replicate and extend Lee (2023), applying it to crypto data and investigating alternative architectures (notably attention/Transformer) for parameter estimation and real-time volatility computation.

Specific objectives and scope

Specific objectives

1. Implement the 2D marked Hawkes model and the **MLE estimator** (baseline).
2. Build the **simulator** (Ogata thinning) and the training-set generator.
3. Reproduce Lee's (2023) **3-layer LSTM**.
4. Adapt the pipeline to **real crypto data**.
5. Design alternative architectures: **GRU, deep LSTM and attention/Transformer**.
6. Compare **accuracy, time and robustness**.
7. Apply to **intraday Hawkes volatility**.

Scope

- **2D symmetric** Hawkes process with exponential decay and marks (jumps).
- **"High"** frequency (filtering at $\Delta t = 0.1$ s), not ultra-high.
- Focus on **parameter estimation** and **volatility**, not trading.

Out of scope

Full limit-order-book modelling; price-direction forecasting; order execution.

Theoretical Background

Marked Hawkes price model

Conditional intensity (2D, exponential decay)

$$\boldsymbol{\lambda}_t = \boldsymbol{\mu} + \int_{(-\infty,t]\times E} \boldsymbol{\alpha} \circ \boldsymbol{b}(t-u)\, \boldsymbol{M}(\mathrm{d}u \times \mathrm{d}z), \quad \boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 & \alpha_2 \\ \alpha_2 & \alpha_1 \end{bmatrix}, \; \boldsymbol{b}(t) = e^{-\beta t}.$$

- μ: baseline intensity; α_1: self-excitation; α_2: mutual excitation; β: decay.
- Moves encoded as **up (1)** and **down (2)**; marks Z_i = jump size.
- **Stability**: branching ratio < 1 (stationary state).

Hawkes volatility

Under stationarity, the standard deviation of the net move has a **closed form** in $\{\mu, \alpha_1, \alpha_2, \beta\}$ — the basis for real-time volatility [Lee & Seo, 2017; Lee, 2024].

Two estimation paths

MLE (traditional)

$$L(T, \boldsymbol{\theta}) = \sum_i \int \log \lambda_i \, \mathrm{d}M_i - \int_0^T \lambda_i \, \mathrm{d}u$$

- Iterative numerical maximisation; accurate but **slow** and initial-value sensitive.

Neural network (Lee, 2023)

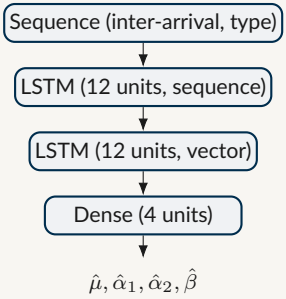
- Generates thousands of **simulated paths** with known parameters.
- Trains a network to **predict the parameters** from the sequence (inter-arrivals + type), with **near-instant** inference.

Lee (2023) result — simulation

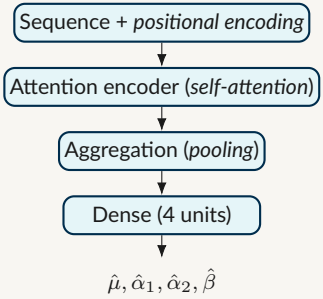
	Network (LSTM)	MLE
MSE	0.0513	0.0417
Time (s)	**0.0120**	1.763

Comparable accuracy; $\sim$**147**$\times$ **faster**.

From LSTM to attention



Baseline — Lee (2023)



Proposal — attention/Transformer

Intuition

Self-attention [Vaswani et al., 2017] directly relates distant events in the sequence — better capturing the long-range memory of jump clustering than the recurrent LSTM [Hochreiter & Schmidhuber, 1997].

Related Work

Literature review — comparative synthesis

Where this work stands

Work	Estimation	Architecture	Market	Volat.	Crypto
Hawkes (1971)	—	—	—	×	×
Lee & Seo (2017)	MLE	—	Equities	✓	×
Bacry et al. (2013)	MLE	—	Equities	∼	×
Mei & Eisner (2017)	Network	RNN (neural Hawkes)	general	×	×
Taddeo et al. (2024)	MLE	—	Crypto	×	✓
Primo (2018)	MLE/expl.	—	Equities (BR)	×	×
Lee (2023) (ref.)	Network	LSTM	Equities (US)	✓	×
This work	Network	**LSTM + attention**	**Crypto**	✓	✓

- Neural-network estimation is still **rare** in financial time series and **novel** at the "crypto + attention + Hawkes volatility" intersection.

Dataset

Dataset — matching the reference paper

What Lee (2023) needs

- **Mid-price** from the NBBO (best bid/ask), sampled every $\Delta t = 0.1$ s.
- On each change: record **time** and **new price** (up/down).
- The minimum **tick size** ϑ scales the price volatility.

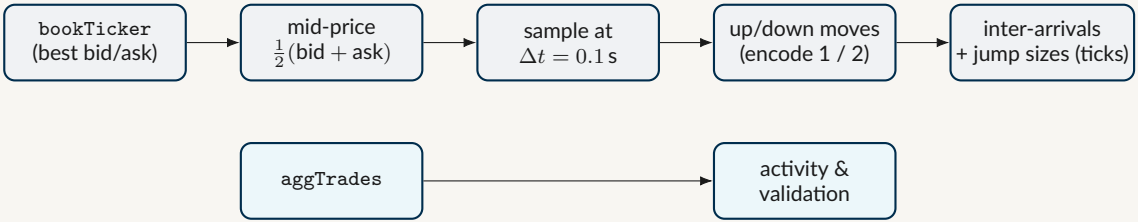
The free crypto analogue

- **Binance Public Data** (`data.binance.vision`) — free, no sign-up.
- `bookTicker`: best bid/ask updates $\Rightarrow$ mid-price (the NBBO analogue).
- `aggTrades`: aggregated trades $\Rightarrow$ activity / validation.
- Assets: **BTCUSDT** and **ETHUSDT** USDT-margined perpetual futures.

`bookTicker` columns

`update_id`, `best_bid_price`, `best_bid_qty`, `best_ask_price`, `best_ask_qty`, `transaction_time`, `event_time` — millisecond timestamps; available since 2019-09-21.

Dataset — from raw ticks to model inputs



Details

- Tick size ϑ from `exchangeInfo` (PRICE_FILTER).
- Intraday seasonality handled via *business time* (re-clocking by average activity).

Period

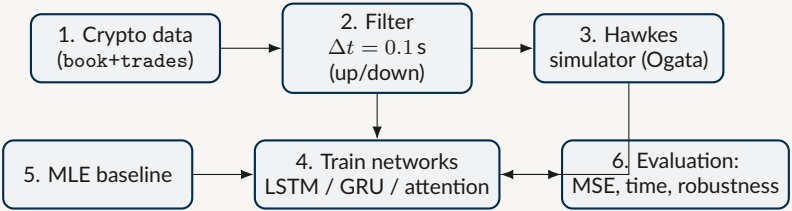
Parameterizable: the source and method are fixed; the exact date range (and whether to include a high-volatility window) is set during TG-2.

Proposal and Methodology

Proposal

Proposed work

Build an end-to-end pipeline for **neural-network Hawkes estimation on cryptocurrencies** and compare architectures, culminating in real-time volatility measurement.



Methodology — details

Data and simulation

- BTCUSDT & ETHUSDT perpetual futures; mid-price filtered at 0.1 s.
- Parameter ranges calibrated by MLE on real windows.
- Simulated paths of $\sim$2000 steps as training data.

Training

- **Adam** optimiser [Kingma & Ba, 2014]; **MSE** loss on the 4 parameters.
- **Softplus** + predicting $\beta - \alpha_1 - \alpha_2$: ensures positivity and branching ratio < 1 [Lee, 2023].

Architectures evaluated

- **3-layer LSTM** (replication).
- **GRU** and **deep LSTM** (variants).
- **Attention encoder / Transformer** (main proposal).

Evaluation

- **Accuracy**: MSE vs. MLE on simulation and real data.
- **Cost**: inference time.
- **Application**: intraday Hawkes volatility vs. **realized volatility** [Andersen et al., 2003].

Success Criteria and Risks

Success criteria

How we will know it worked

1. **(C1) Replication:** qualitatively reproduce Lee (2023) — MSE of the same order as MLE and inference time $< 1\%$ of MLE.
2. **(C2) Real pipeline:** end-to-end estimation on real crypto data, respecting model constraints (positivity, branching ratio < 1).
3. **(C3) Architectural contribution:** at least one attention-based architecture with MSE $\leq$ the LSTM baseline.
4. **(C4) Volatility:** an intraday volatility series that is stable and consistent with the realized-volatility benchmark.

Guaranteed minimum deliverable

C1 + C2 already constitute a complete thesis; C3 and C4 are the higher-contribution extensions.

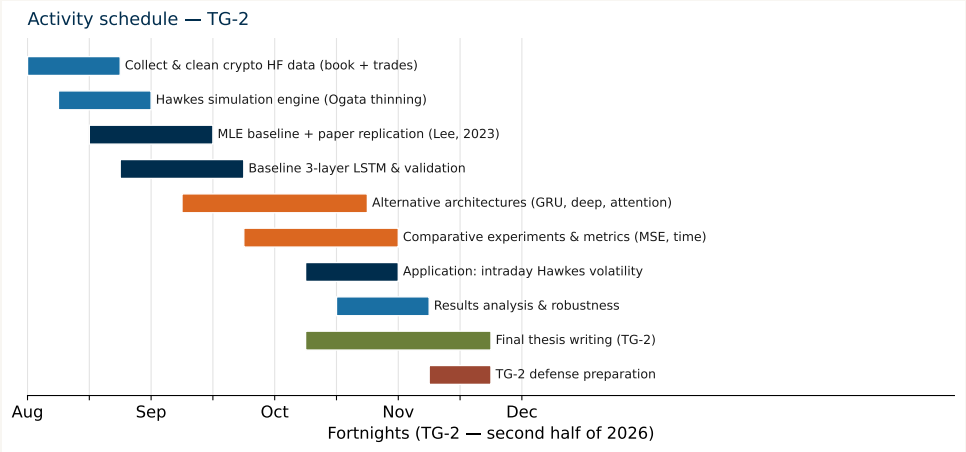
Main risks and mitigations

Risk	P×I	Mitigation
Microstructure noise / crypto data quality	M×H	Filtering at $0.1\,\text{s}$; public `bookTicker`/`aggTrades`; cleaning and validation.
Model mis-specification (*subdue* effect, negative α)	M×M	Softplus + predict $\beta - \alpha_1 - \alpha_2$; hybrid training (simulation + MLE).
Non-stationarity (24/7 market, regimes)	H×M	Window/segment estimation; per-regime analysis.
Transformer training cost / instability	M×M	Start simple; regularisation; LSTM baseline guarantees delivery.
Scope creep	M×H	C1+C2 as the core; attention and volatility as increments.

P = probability, I = impact; L/M/H = low/medium/high.

Schedule and Next Steps

Activity schedule (TG-2)



The core (data → simulator → MLE → LSTM) is delivered by mid-September.

Next steps

From the proposal (TG-1) to TG-2

- Finish collecting and cleaning the high-frequency cryptocurrency data.
- Implement the Hawkes simulator and the MLE baseline.
- Reproduce Lee's (2023) LSTM and validate against the published results.
- Design and train the alternative architectures (GRU, deep, attention).
- Run the comparative experiments and the volatility application.
- Write the final thesis and prepare the TG-2 defense.

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Thank you!

Questions?

Appendix — Mathematical Background

A.1 — Branching ratio, stationarity and memory

Mean intensity

From $\phi(t) = \mu + \int_{-\infty}^{t} \Phi(t-u)\,\mathrm{d}N(u)$, take the stationary expectation ($\mathbb{E}[\mathrm{d}N] = \bar{\phi}\,\mathrm{d}u$):

$$\bar{\phi} = \mu + \bar{\phi}\int_0^\infty \Phi(s)\,\mathrm{d}s = \mu + g\,\bar{\phi},$$

with $g := \int_0^\infty \Phi(s)\,\mathrm{d}s$. Hence

$$\boxed{\bar{\phi} = \frac{\mu}{1-g}} \quad (g < 1).$$

Branching interpretation

Each event spawns offspring with intensity Φ; the expected number of direct children is

$$\mathbb{E}[\#\text{children}] = \int_0^\infty \Phi(s)\,\mathrm{d}s = g.$$

Mean family size (geometric series):

$$\bar{S} = 1 + g + g^2 + \cdots = \tfrac{1}{1-g}.$$

Effective memory ($\Phi = \alpha e^{-\beta t}$): $\tau_{\text{eff}} \approx \frac{1}{\beta} \cdot \frac{1}{1-g}$ — diverges as $g \to 1$.

A.2 — From the scalar definition to the 2D marked intensity

Three steps

1. **Multivariate (2D):** types up (1)/down (2), each with λ_i; the kernel becomes a matrix $\lambda_i(t) = \mu_i + \sum_j \int_{-\infty}^{t} \Phi_{ij}(t-u)\,\mathrm{d}N_j(u)$.
2. **Symmetry + exponential:** $\Phi_{ij}(t) = \alpha_{ij} e^{-\beta t}$, $\boldsymbol{\alpha} = \left[\begin{smallmatrix} \alpha_1 & \alpha_2 \\ \alpha_2 & \alpha_1 \end{smallmatrix}\right] \Rightarrow \boldsymbol{\Phi} = \boldsymbol{\alpha} \circ \boldsymbol{b}(t)$.
3. **Marks:** $\boldsymbol{M}(\mathrm{d}u \times \mathrm{d}z)$ records (time, jump Z); $\mathrm{d}N = \int_z \boldsymbol{M}$. The intensity ignores z; the *price* uses Z (hence "marked").

Dropping the 2nd dimension and marks, with $\Phi = \alpha e^{-\beta t}$, recovers the Context definition.

Stationarity → what the network predicts

Spectral radius of $\int_0^\infty \boldsymbol{\alpha} \circ \boldsymbol{b}\,\mathrm{d}t = \boldsymbol{\alpha}/\beta < 1$; eigenvalues $(\alpha_1 \pm \alpha_2)/\beta$ give $(\alpha_1 + \alpha_2)/\beta < 1 \iff \beta - \alpha_1 - \alpha_2 > 0$ — exactly the softplus target in Lee (2023).

A.3 — The maximum-likelihood objective

Bin argument

Over $[0, T]$ split into bins $\mathrm{d}t$: $P(\text{event}) \approx \lambda\,\mathrm{d}t$, $P(\text{none}) \approx e^{-\lambda\,\mathrm{d}t}$. The product over all bins gives

$$L = \Big[\prod_k \lambda(t_k)\Big]\, e^{-\int_0^T \lambda(u)\,\mathrm{d}u} \;\Rightarrow\; \log L = \sum_k \log \lambda(t_k) - \int_0^T \lambda(u)\,\mathrm{d}u.$$

Multivariate marked form

$$L(T, \boldsymbol{\theta}) = \sum_{i=1}^{2} \Big[\sum_{t_{i,n}} \log \lambda_i(t_{i,n}) - \int_0^T \lambda_i(u)\,\mathrm{d}u\Big],$$

the first sum running over the type-i event times. Substitute $\lambda_i(t) = \mu + \sum \alpha e^{-\beta(t-t_i)}$ and maximise over $\{\mu, \alpha_1, \alpha_2, \beta\}$ — an iterative, numerical step, replaced here by the network's learned inference.